

Circles (Set B) — MCQ Worksheet

Mathematics • Chapter 10 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. Tangent to a circle is perpendicular to:

- (A) The chord
- (B) The diameter only
- (C) The radius at point of contact
- (D) Another tangent

Q2. From an external point, number of tangents to a circle is:

- (A) 1
- (B) 2
- (C) 3
- (D) Infinitely many

Q3. Two tangents from an external point are:

- (A) Always perpendicular
- (B) Equal in length
- (C) Unequal
- (D) Always parallel

Q4. Length of tangent from a point 13 cm away from centre of circle of radius 5 cm:

- (A) 8 cm
- (B) 12 cm
- (C) 10 cm
- (D) 5 cm

Q5. If radii of two concentric circles are 4 cm and 5 cm, then length of chord of larger circle which touches smaller circle is:

- (A) 3 cm
- (B) 6 cm
- (C) 7 cm
- (D) 8 cm

Q6. Number of tangents at a point on a circle is:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q7. If two tangents drawn from external point P to a circle subtend 70° at centre, $\angle APB =$

- (A) 70°
- (B) 110°
- (C) 120°
- (D) 90°

Q8. Distance between centre and tangent of a circle equals:

- (A) Diameter
- (B) Half of chord
- (C) Radius
- (D) Zero

Q9. The lengths of tangents drawn from an external point P to a circle with centre O are equal because:

- (A) Triangle OAP $\cong$ Triangle OBP (RHS criterion)
- (B) Both lines coincide
- (C) Tangents always cross
- (D) Lengths are unrelated

Q10. A circle is inscribed in a triangle. Number of tangents from each vertex is:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q11. If two tangents from external point P are inclined at 60° , then OP/OA (where O is centre, A is point of contact) =

- (A) 2
- (B) 1
- (C) $\sqrt{3}$
- (D) $\sqrt{2}$

Q12. Length of common tangent to two externally tangent circles of radii r and R is:

- (A) $\sqrt{2rR}$
- (B) $2\sqrt{rR}$
- (C) $r + R$
- (D) $R - r$

Q13. In figure, PA and PB are tangents to a circle from P. If $\angle APB = 80^\circ$, then $\angle OAB =$

- (A) 40°
- (B) 50°
- (C) 60°
- (D) 30°

Q14. Tangent at any point of a circle and the radius through the point form an angle of:

- (A) 60°
- (B) 90°

(C) 30°

(D) 45°

Q15. Number of common tangents to two non-overlapping circles is:

(A) 1

(B) 2

(C) 3

(D) 4

Q16. In figure, PT is tangent to circle with centre O. If $OT = 13$ cm and $PT = 12$ cm, then radius is:

(A) 5 cm

(B) 6 cm

(C) 7 cm

(D) 12 cm

Q17. A circle is inscribed in a quadrilateral ABCD touching AB, BC, CD, DA at P, Q, R, S. If $AB = 6$, $BC = 7$, $CD = 4$, then $DA =$

(A) 3

(B) 5

(C) 9

(D) 4

Q18. Two concentric circles have radii 5 cm and 3 cm. The length of chord of bigger circle which is tangent to smaller is:

(A) 4 cm

(B) 6 cm

(C) 8 cm

(D) 10 cm

Q19. AB is diameter and CD is chord of a circle such that $AB = 26$ cm, $CD = 10$ cm. If $CD \perp AB$ at E (and $AE > EB$), distance from centre to chord CD is:

(A) 12 cm

(B) 13 cm

(C) 14 cm

(D) 10 cm

Q20. In an isosceles triangle ABC with $AB = AC$, a circle through B and C touches AB at B. The line through C parallel to AB cuts the circle again at D. Then ABCD is:

(A) Trapezium

(B) Isosceles trapezium

(C) Parallelogram

(D) Rectangle

Q21. From a point P, two tangents PA and PB are drawn to a circle of radius r centred at O. If $OP = 2r$, then $\angle APB =$

(A) 30°

(B) 45°

(C) 60°

(D) 90°

Q22. A quadrilateral ABCD is drawn to circumscribe a circle. $AB = 6$ cm, $BC = 7$ cm, $CD = 4$ cm. Find AD.

(A) 3 cm

(B) 5 cm

(C) 9 cm

(D) 8 cm

Q23. The angle in alternate segment theorem states: the angle between a tangent and chord equals:

(A) Angle subtended by chord in alternate segment

(B) Twice the angle at centre

(C) Half the chord length

(D) Zero

Q24. If tangents PA and PB from external point P to a circle with centre O are inclined at 60° , what is the length of OP if $OA = 2$ cm?

(A) 2 cm

(B) 4 cm

(C) $2\sqrt{3}$ cm

(D) $4\sqrt{3}$ cm

Q25. If a circle touches the side BC of triangle ABC at P and extension of AB, AC at Q, R respectively, then $AQ =$

(A) $\frac{1}{2}$ perimeter of ABC

(B) Twice perimeter

(C) Perimeter

(D) Half side BC